

192SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.Sandeep(192525095)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ____Sandeep M_____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1)Answer

Auto-Scaling

Auto-scaling automatically increases or decreases the number of cloud servers based on demand. During a seasonal sale, when thousands of customers visit the website at the same time, the cloud platform launches additional virtual servers to handle the extra traffic. Once the sale ends and traffic returns to normal, the extra servers are removed. This ensures efficient use of resources while reducing unnecessary costs.

Load Balancing

A load balancer distributes incoming user requests evenly across multiple servers. Instead of sending all traffic to one server, it routes requests to servers with available capacity. If one server becomes overloaded or fails, the load balancer automatically redirects traffic to healthy servers. This prevents crashes and ensures continuous service.

Reducing Downtime and Improving User Experience

The combination of auto-scaling and load balancing offers several benefits:

- Prevents server overload during traffic spikes.
- Reduces website downtime by automatically replacing failed or overloaded instances.
- Maintains fast page loading and smooth checkout processes.
- Provides a seamless shopping experience, increasing customer satisfaction and reducing abandoned carts.

Role of Monitoring Tools

Monitoring tools continuously track system performance using metrics such as CPU usage, memory utilization, response time, network traffic, and error rates. These tools:

- Detect unusual traffic patterns or performance issues.
- Trigger auto-scaling when predefined thresholds are reached.
- Alert administrators about system failures or security concerns.
- Provide performance reports that help optimize future scaling strategies.

Real-World E-commerce Scenario

Consider an online retailer such as Amazon during a major shopping event like Prime Day. Millions of customers visit the website simultaneously to purchase discounted products. Cloud auto-scaling automatically adds more servers as traffic increases, while load balancers distribute customer requests across these servers to prevent overload. Monitoring tools continuously observe server health and application performance, triggering additional scaling when needed and notifying engineers of any issues. As a result, customers experience fast browsing, reliable payments, and uninterrupted service despite the massive surge in demand.

Conclusion

Cloud auto-scaling, load balancing, and monitoring tools work together to handle unpredictable traffic efficiently. They ensure high availability, minimize downtime, optimize resource usage, and provide a smooth user experience, making them essential technologies for modern e-commerce platforms during peak shopping events.

2)Answer

Cloud Storage for Secure Backup Data

Cloud storage is an ideal solution for organizations that need to store large volumes of backup data securely and reliably. It provides high durability, redundancy, security, and cost savings compared to traditional storage systems.

1. Durability and Data Redundancy

Cloud providers ensure that backup data is safe by storing multiple copies across different servers and data centers.

- **Durability:** Data is designed to remain intact even if hardware fails.
- **Data Redundancy:** Multiple copies of the same data are stored in different physical locations.
- If one server or data center fails, another copy is immediately available.
- Automatic replication and regular integrity checks ensure data is not lost or corrupted.

2. Encryption and Access Control

Sensitive backup data is protected through strong security measures.

Encryption

- Encryption at Rest: Data is encrypted while stored in the cloud.
- Encryption in Transit: Data is encrypted while being transferred over the internet using secure protocols such as TLS.
- Even if unauthorized users access the storage, encrypted data cannot be read without the encryption key.

Access Control

- Identity and Access Management (IAM) restricts who can access backup data.
- Multi-Factor Authentication (MFA) adds an extra layer of security.
- Role-Based Access Control (RBAC) ensures users only access the data required for their role.
- Audit logs record all access attempts for security monitoring.

3. Cost Advantages over Traditional Storage

Cloud storage is more cost-effective than maintaining on-premises backup infrastructure.

- No need to purchase expensive storage hardware.
- Eliminates maintenance and hardware replacement costs.
- Pay only for the storage actually used (pay-as-you-go model).
- Easily scale storage up or down without additional infrastructure investment.
- Reduced power, cooling, and physical space requirements.

4. Disaster Recovery Use Case

A company stores daily backups of its customer database in cloud storage.

One day, the company's primary data center is damaged due to flooding or a fire. The local backup servers are also affected. Since backup copies are stored securely in the cloud across multiple regions, the organization quickly restores its systems from the cloud backups to a new server. Business operations resume with minimal downtime and very little data loss, ensuring business continuity.

Conclusion

Cloud storage provides high durability, automatic redundancy, strong encryption, and strict access control to protect backup data. It also reduces infrastructure costs through its pay-as-

you-go pricing model. In disaster recovery situations, cloud backups enable organizations to restore critical data quickly and continue operations with minimal downtime.

3)Answer

Global Deployment Using CDN and Cloud Services

A mobile app company deploying its application worldwide must ensure fast content delivery and low response times. Content Delivery Networks (CDNs), edge servers, and global cloud infrastructure help achieve high performance and availability for users across different regions.

1. How CDNs Improve Performance and Reduce Latency

A Content Delivery Network (CDN) is a network of distributed servers located in multiple geographic regions.

- Frequently accessed content (images, videos, app updates, CSS, JavaScript, etc.) is cached on servers close to users.
- User requests are served from the nearest CDN server instead of the main server.
- This reduces the distance data travels, resulting in lower latency.
- CDNs reduce the load on the origin server, improving application performance during high traffic.

2. How Edge Servers Enhance User Experience

Edge servers are located near end users and process requests locally.

Benefits include:

- Faster loading of web pages and mobile apps.
- Reduced buffering during video streaming.
- Lower response time in online gaming.
- Improved reliability even during traffic spikes.
- Better performance for users in different countries and regions.

3. Cloud Support for Global Availability

Cloud providers operate multiple data centers worldwide to ensure applications remain available.

They provide:

- Multi-region deployment to host applications in different geographic locations.
- Load balancing to distribute traffic across regions.
- Automatic failover to redirect users if one region experiences an outage.
- Auto-scaling to add or remove servers based on demand.
- 24/7 monitoring to maintain high availability and quickly detect issues.

These features ensure users can access the application with minimal downtime, regardless of their location.

4. Real-Time Streaming or Gaming Example

Consider a global online gaming platform such as PUBG Mobile or Free Fire MAX.

- Players from Asia, Europe, and North America connect simultaneously.
- CDN edge servers deliver game updates and downloadable content from nearby locations.
- Game servers hosted in multiple cloud regions reduce network delay (ping).
- Load balancers distribute player connections across available servers.
- Auto-scaling automatically launches additional servers during tournaments or special events.
- If one cloud region fails, traffic is redirected to another region without interrupting gameplay.

As a result, players experience low latency, faster downloads, smooth gameplay, and uninterrupted service.

Conclusion

CDNs, edge servers, and global cloud infrastructure work together to deliver content quickly and reliably worldwide. They reduce latency, improve application speed, ensure high availability, and provide an excellent user experience, making them essential for modern streaming and gaming applications.

4)Answer

Cloud Security, Compliance, and Data Sovereignty in Financial Institutions

Financial institutions handle highly sensitive customer and transaction data. Cloud providers offer regulatory compliance, auditing, encryption, logging, and monitoring to help organizations meet security and legal requirements.

1. Cloud Support for Regulatory Compliance and Auditing

Cloud providers comply with international and industry security standards to help organizations meet legal and regulatory requirements.

They support compliance by:

- Maintaining certifications such as ISO 27001, SOC 2, and PCI DSS for payment security.
- Providing audit reports and compliance documentation.
- Offering identity and access management (IAM) to control user permissions.
- Maintaining detailed audit trails of all user activities.
- Supporting regular security assessments and compliance checks.

These features help organizations demonstrate compliance with financial regulations.

2. Role of Logging, Monitoring, and Encryption

Logging

- Records every login, transaction, file access, and configuration change.
- Creates an audit trail for investigations and regulatory audits.
- Helps detect unauthorized activities.

Monitoring

- Continuously tracks system performance, user activity, and security events.
- Detects unusual behavior such as unauthorized login attempts or suspicious transactions.
- Sends real-time alerts so administrators can respond quickly.

Encryption

- Encryption at Rest: Protects stored financial records.
- Encryption in Transit: Protects data while it is transmitted over networks.
- Even if data is intercepted, it cannot be read without the encryption key.

Together, these measures protect sensitive financial information from cyber threats.

3. Ensuring Data Sovereignty

Data sovereignty means that data must remain within a specific country or region to comply with local laws.

A company can ensure this by:

- Selecting cloud regions located within the required country.
- Configuring storage and backups to remain in approved geographic locations.
- Preventing unauthorized cross-border data transfers.
- Using regional disaster recovery sites that comply with local regulations.
- Regularly auditing data locations to verify compliance.

This ensures customer data is stored and processed according to national regulations.

4. Example: Sensitive Financial Transactions

A bank processes online fund transfers and credit card payments.

- Customer account details and transaction records are encrypted during transmission and while stored.
- Every login, transaction, and administrative action is logged for auditing.
- Monitoring systems detect unusual activities, such as multiple failed login attempts or unusually large transfers, and immediately alert the security team.
- All customer data is stored in cloud data centers located within the country to satisfy data sovereignty laws.
- During regulatory audits, the bank provides audit logs and compliance reports to demonstrate that all transactions were securely processed and protected.

Conclusion

Cloud providers help financial institutions achieve regulatory compliance through certifications, auditing tools, secure logging, continuous monitoring, and strong encryption. By choosing appropriate cloud regions and controlling where data is stored, organizations can also meet data sovereignty requirements while ensuring secure and reliable financial services.

5)Answer

1. How Kubernetes Simplifies Deployment and Scaling

Kubernetes is a container orchestration platform that automates the deployment, management, and scaling of containerized applications.

It provides:

- **Automatic Deployment:** Deploys containers across multiple servers.
- **Auto-Scaling:** Increases or decreases the number of containers based on traffic.
- **Load Balancing:** Distributes requests evenly among running containers.
- **Self-Healing:** Restarts failed containers and replaces unhealthy ones automatically.
- **Rolling Updates:** Updates applications without downtime and supports easy rollback if issues occur.

These features reduce manual effort and improve application reliability.

2. Portability Across Different Cloud Environments

Containers package the application together with all required libraries and dependencies.

This enables developers to:

- Run the same application on a laptop, private cloud, or public cloud without modification.
- Avoid compatibility issues between development, testing, and production environments.
- Easily migrate applications between cloud providers (such as AWS, Azure, and Google Cloud).
- Support hybrid and multi-cloud deployments.

This portability makes containers an excellent choice for modern cloud applications.

3. Example: Microservices-Based Application

Consider an online shopping (e-commerce) application built using microservices.

The application is divided into independent services such as:

- User Service – Manages user registration and login.
- Product Service – Handles product catalog information.
- Order Service – Processes customer orders.
- Payment Service – Manages online payments.
- Notification Service – Sends email or SMS updates.

Each service runs inside its own container.

When a large sale increases customer traffic:

- Kubernetes automatically scales the Product and Order services by creating additional container instances.
- Load balancing distributes customer requests among these containers.
- If a container fails, Kubernetes automatically restarts or replaces it without affecting the other services.

Conclusion

Containers are lighter, faster, and more resource-efficient than virtual machines. Kubernetes simplifies container deployment through automation, scaling, load balancing, and self-healing. Containerization also provides excellent portability across different cloud environments, making it ideal for microservices-based applications such as e-commerce platforms, streaming services, and banking systems.

